DARSHAN S

AI/ML Engineer
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GitHub Portfolio LinkedIn

Professional Summary

AI & ML undergraduate with experience building computer vision systems for geospatial property analysis and document-grounded RAG applications. Skilled in Python, FastAPI, React, semantic caching, intelligent LLM routing, model evaluation, and full-stack development. Focused on developing practical AI systems with measurable cost optimization, reliable retrieval, and observability.

Education

M S Ramaiah University of Applied Sciences — *In Progress*
B.Tech, Artificial Intelligence & Machine Learning — CGPA: 8.0

Christ Junior College — *2023*
Pre-University Course, KSEAB, Karnataka — 83.6%

Cambridge Public School — *2021*
10th Boards, CISCE — 92.8%

Experience

MPRO9 — *Jul. 2026 – Oct. 2026*
AI Intern – Project Development & Implementation — Mysore, India

- Built GeoVision AI for property identification and analysis using geospatial and visual data.
- Developed YOLO-based object detection workflows for analyzing 360-degree imagery.
- Integrated visual data with GPS information using Insta360 X4 imagery and u-blox GPS modules.
- Contributed to data preprocessing, model development, testing, and optimization.

Personal Projects

Cost-Aware RAG – Semantic Caching & Intelligent Model Routing
Python, FastAPI, React, TypeScript, SQLite, Groq, Sentence Transformers

- Built a document-grounded RAG pipeline with PDF/TXT ingestion, page-aware chunking, vector retrieval, source citations, and insufficient-evidence handling.
- Implemented evidence-aware semantic caching and complexity-based LLM routing, reducing estimated inference cost by **59.2% across 38 requests** in offline evaluation.
- Created isolated temporary visitor workspaces using server-issued tokens and in-memory SQLite, with session-scoped documents, retrieval, caching, and telemetry.

Smart Traffic Vehicle Counter
Python, YOLO11, ByteTrack, OpenCV, Streamlit

- Developed a vehicle detection, tracking, classification, and counting system with direction and speed estimation.
- Combined YOLO11, ByteTrack, trajectory analysis, and line-crossing logic in a Streamlit analytics dashboard.

AI Health Tracker
React.js, Node.js, Express.js, MongoDB, Google Gemini API, JWT

- Built a full-stack healthcare platform for health-record management and personalized AI-generated insights.
- Implemented JWT authentication, REST APIs, MongoDB persistence, and a responsive React dashboard.

Spam Mail Classifier
Python, Pandas, NumPy, TF-IDF, Logistic Regression

- Developed an NLP spam classifier achieving **98% classification accuracy**.
- Applied TF-IDF and Logistic Regression, evaluating results with precision, recall, F1-score, and a confusion matrix.

Technical Skills

Languages: Python, JavaScript, Java, C
Frontend: React.js, TypeScript, HTML5, CSS3, Tailwind CSS
Backend: FastAPI, Node.js, Express.js, REST APIs
AI/ML: RAG, Semantic Caching, LLM Routing, Sentence Transformers, TensorFlow, Scikit-learn, YOLO, Computer Vision, NLP, Deep Learning, LangChain, LangGraph, Google Gemini API
Databases: SQLite, MongoDB, SQL
Tools: Git, GitHub, Postman, VS Code, Jupyter Notebook, Pytest, OpenCV, Streamlit
Hardware: Insta360 X4, u-blox GPS
Coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Operating Systems, Computer Networks

Certifications

- Oracle Certified: AI Vector Search Certified Professional
- AI Fluency Framework and Foundations – Anthropic
- AI Literacy – IBM
- AI Fundamentals with IBM SkillBuild – Cisco
- Data Analytics Essentials – Cisco
- Prompt Design in Vertex AI – Google
- Generative AI Foundations – upGrad in collaboration with Microsoft